A Piecewise-Affine Border-Collision Hénon Pilot

Anonymous Route-A report

Abstract

We freeze a two-branch piecewise-affine Hénon map with a border at $x = 0$. Exact affine returns verify every binary itinerary through period eight and give 71 primitive necklaces. A separately frozen branch weight produces an exact finite weighted transfer determinant. The construction is a switching pilot only: it does not assert a global Markov partition or an analytic Fredholm operator.

1 Frozen switching map

For $s \in \{0, 1\}$ let

$$P_s(x, y) = (-5x + c_s - y, x), \qquad (c_0, c_1) = (-2, 2),$$

with domains $x < 0$ and $x > 0$. The border $x = 0$ is excluded rather than assigned after counting. Both branch derivatives equal

$$B = \begin{pmatrix} -5 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \det B = 1.$$

The affine translation is retained in the orbit check, so the calculation is not an averaged transition matrix.

2 Exact finite ledger

For each binary word w of length at most eight we solve $(I - B^n)q = t_w$ over the rationals and iterate the resulting point. Every word has a unique non-border point with the declared itinerary. Cyclic canonicalization gives the primitive counts

$$(2, 1, 2, 3, 6, 9, 18, 30),$$

for periods one through eight, totaling 71. The identity

$$t_n = \sum_{d|n} dp_d$$

is therefore a finite exact necklace relation.

To expose branch chronology without fitting a parameter, freeze $\rho_0 = 1/2$ and $\rho_1 = 2/3$ and put the destination-weighted block $\rho_j B$ on every allowed transition $i \to j$. The resulting four-dimensional screening matrix A has

$$\det(I - zA) = \frac{49z^2 + 210z + 36}{36}, \qquad \det(I - zA_{\text{unweighted}}) = 1 + 10z + 4z^2.$$

The weighted trace prefix begins

$$-\frac{35}{6}, \quad \frac{1127}{36}, \quad -\frac{18865}{108}, \quad \frac{1265327}{1296}.$$

Proposition 1. *The affine-return ledger, primitive quotient, determinant, and trace prefix are reproduced by an independent implementation and a SymPy calculation.*

Proof. The checker reconstructs the affine return and branch inequalities without calling producer functions. The symbolic script independently expands the four-by-four determinant and powers. □

3 Route-A boundary

The finite switching certificate earns ‘A1_PARTIAL_CERTIFIED‘ and ‘A2_CERTIFIED_PREFIX‘. It does not prove that the border map has a complete geometric coding, that boundary orbits are absent at all periods, or that the weighted matrix is a Fredholm determinant. The package makes no arithmetic or Route-B claim. All evidence is protected by the ‘NO_BAD_EULER_OR_ROOT_NUMBER‘ scope firewall.